

INTELLIGENT & SAFETY NAVIGATION PLATFORM

Presented by
K.Harini

2520040011

Department of Electronics Communication and Engeneering (ECE)

Guide: Dr. Sudhakar



Contents



Abstract / Proposal

Overview of the web-based navigation and safety platform designed to assist users in navigating large environments like college campuses, office complexes, and residential buildings.



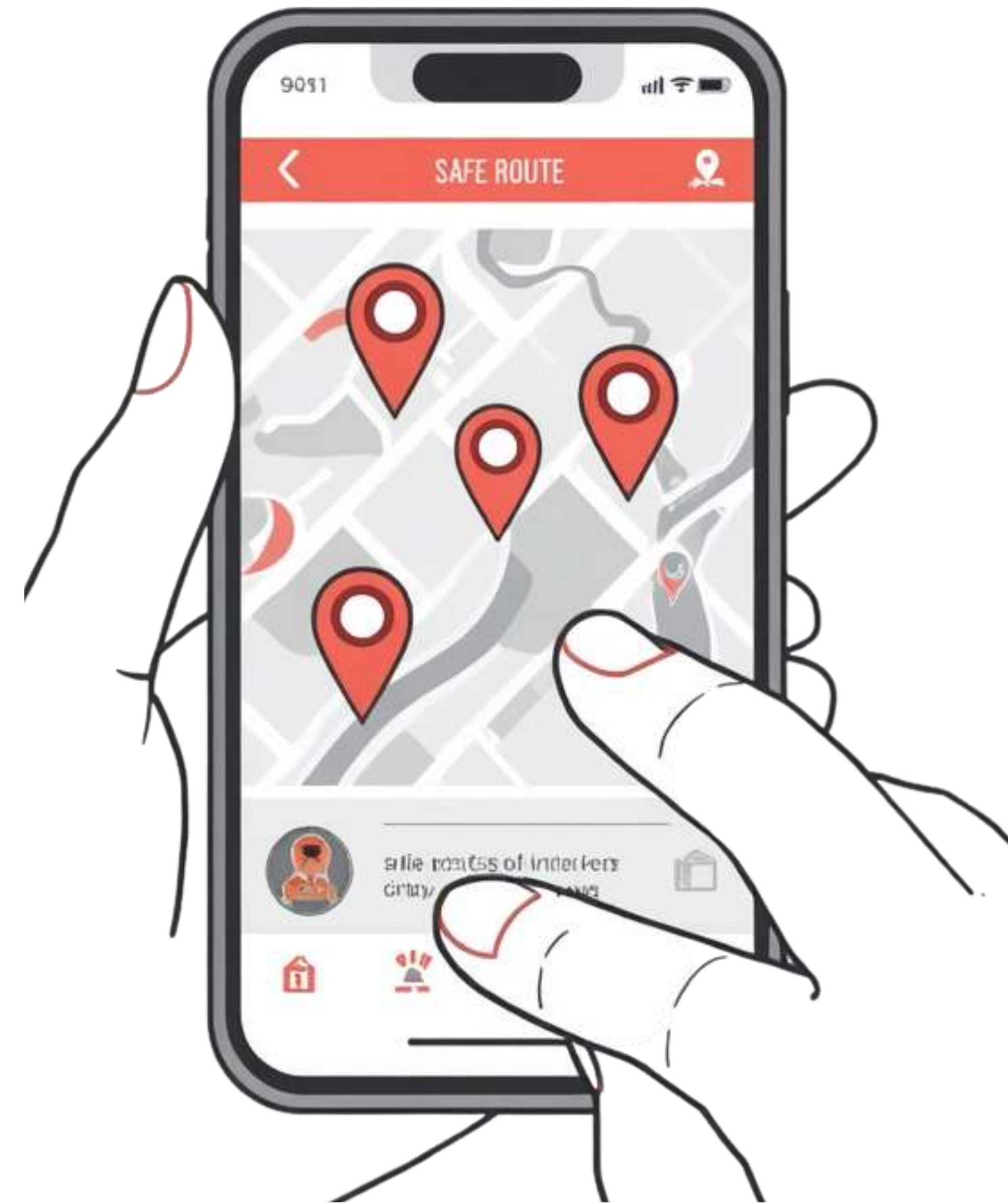
Hardware & Software Requirements

Detailed specifications of the technical requirements including hardware components and software tools needed to develop and deploy the navigation platform.



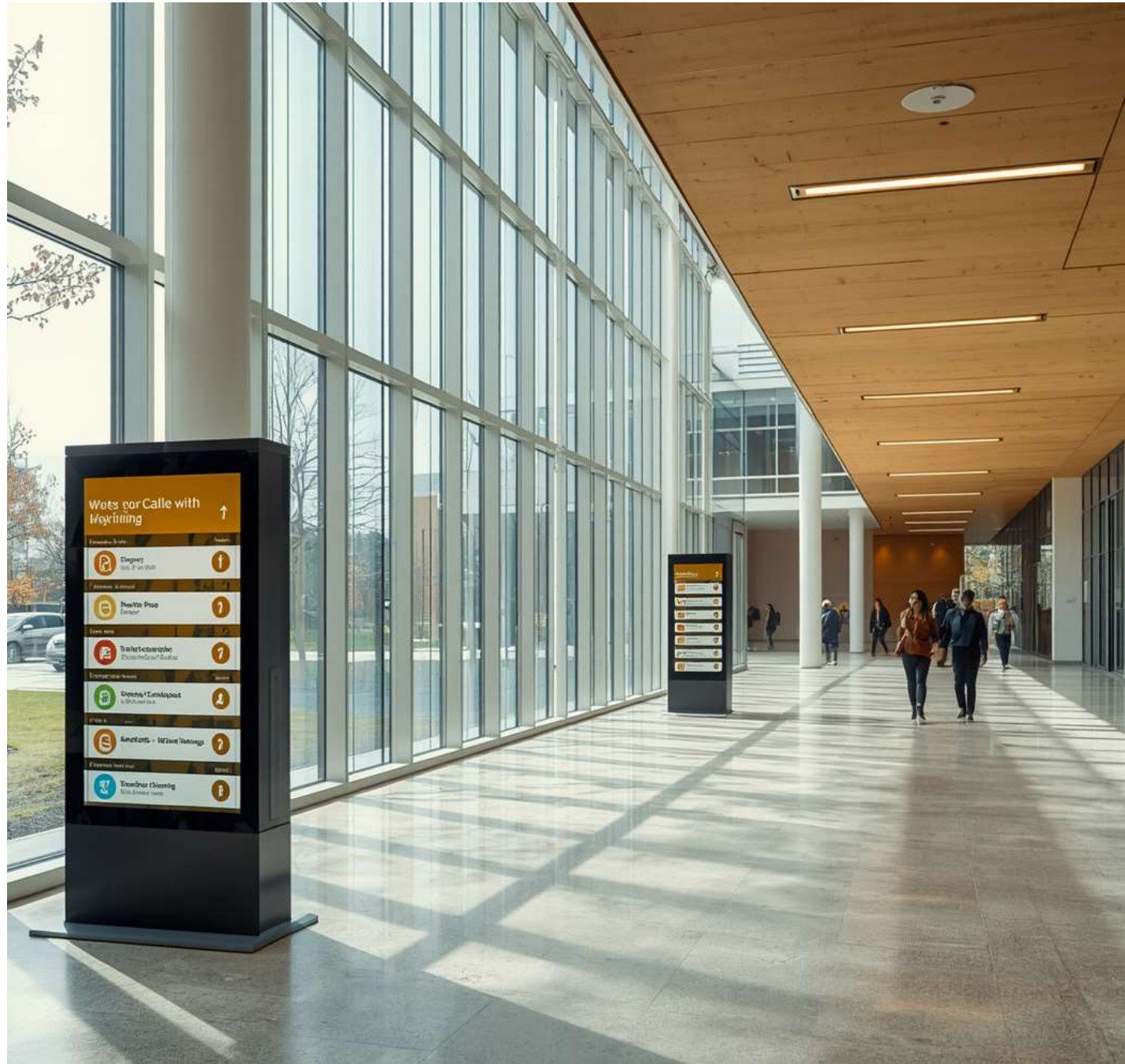
Methodology & System Architecture

Existing systems analysis, proposed system design, and complete architectural overview of the intelligent navigation solution.



Intelligent
Navigation &
Safety Platform

Abstract/Proposal



- Safe Path is a web-based application designed to assist users in navigating large environments such as college campuses, office complexes, and residential buildings.
- Many individuals, especially newcomers and visitors, face difficulty locating specific blocks, departments, or facilities, leading to confusion and time loss.
- During emergency situations such as fire accidents or blocked pathways, the lack of proper guidance increases risk and delays safe evacuation.
- So, our project safe path addresses these challenges through intelligent routing and real-time safety alerts.

Hardware Requirements

Server Infrastructure

Dedicated web server with minimum 16GB RAM, quad-core processor, and 500GB SSD storage for hosting the navigation platform. Cloud-based deployment recommended using AWS or Azure for scalability and reliability. Database server with MySQL or PostgreSQL for storing building maps, user data, and real-time navigation logs.

Network & Display Devices

Wi-Fi access points for indoor positioning, digital signage displays at key locations, and mobile devices (smartphones/tablets) for end-user navigation access.



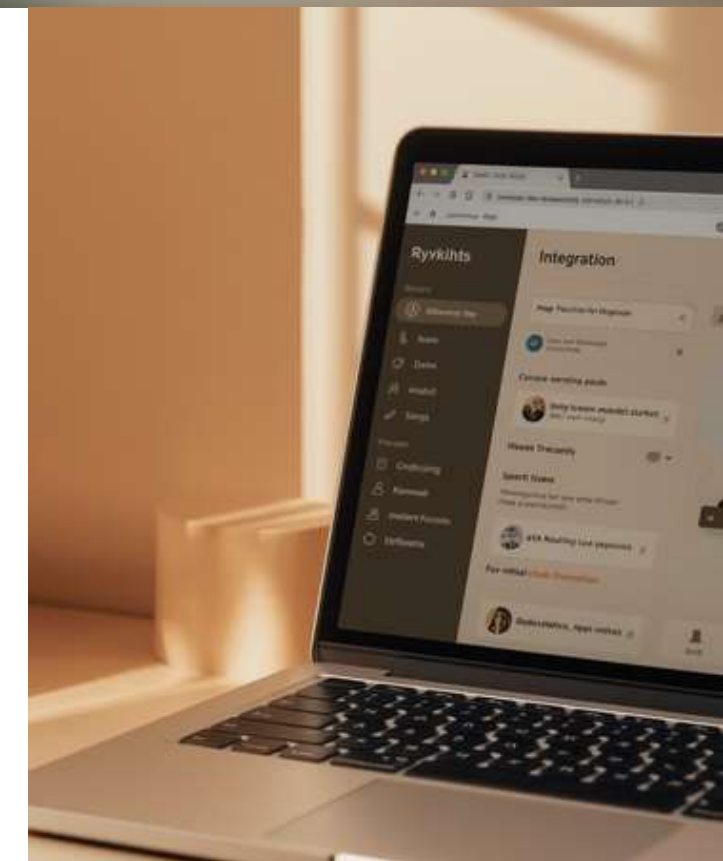
Software Requirements

Web Technologies & Database Systems

- Frontend: HTML5, CSS3, JavaScript, React.js or Angular for responsive UI development
- Backend: Node.js with Express.js or Python with Django/Flask framework
- Database: MySQL or PostgreSQL for relational data storage
- NoSQL: MongoDB for flexible document storage and real-time data handling
- Cloud Platform: AWS, Google Cloud, or Microsoft Azure for hosting and scalability

APIs & Communication Protocols

- Google Maps API or OpenStreetMap for navigation and routing
- WebSocket protocol for real-time emergency alerts
- REST APIs for client-server communication



Existing Systems

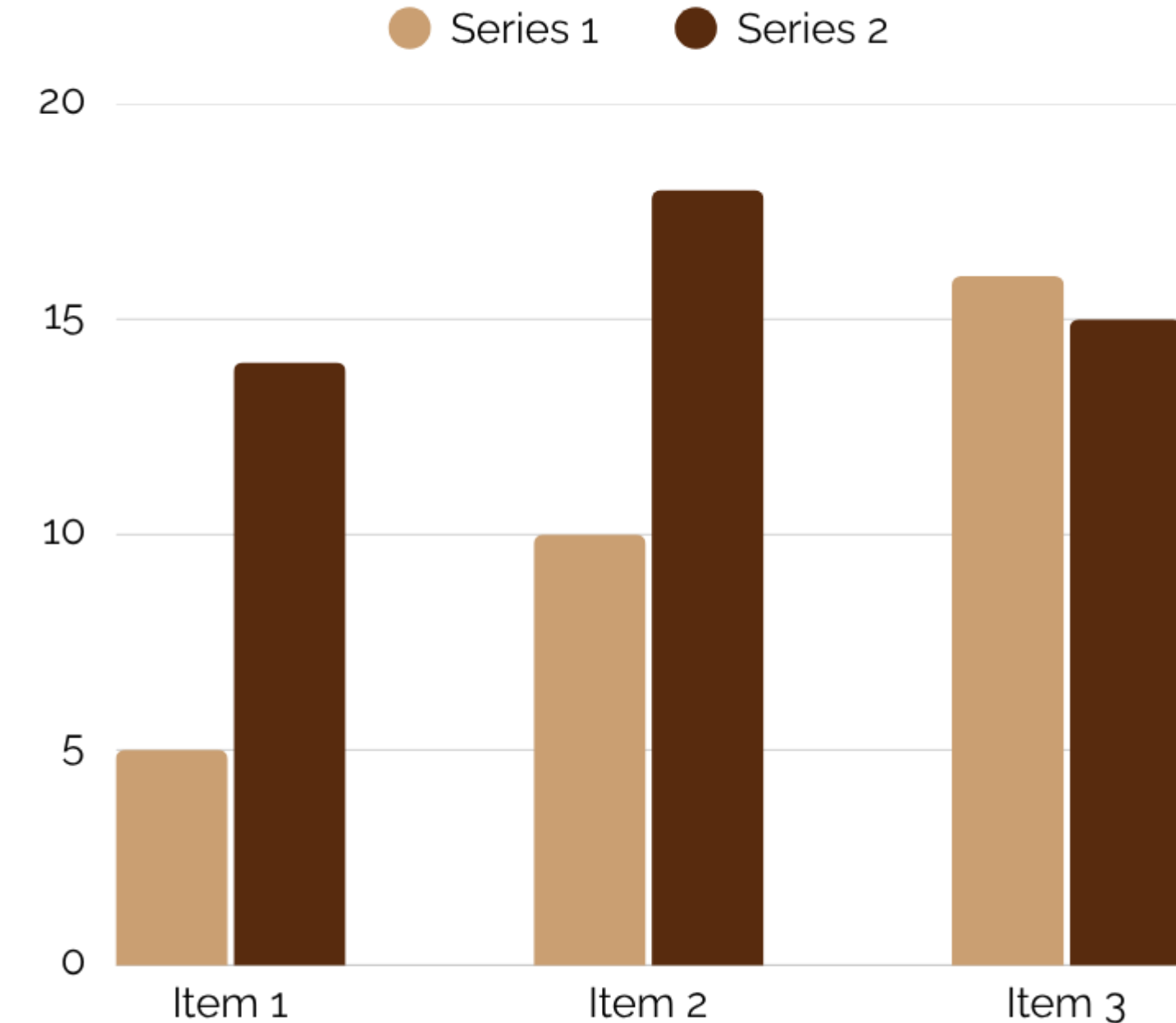
- Current navigation solutions rely heavily on static signage, printed maps, and basic digital directories.
- These systems lack real-time updates and cannot adapt to changing conditions. Users must manually interpret complex floor plans, leading to confusion in unfamiliar environments.



Key limitations include no emergency rerouting capabilities, inability to handle blocked pathways, poor accessibility features, and lack of personalized guidance. During emergencies like fires, these static systems become dangerous as they cannot redirect users away from hazards.

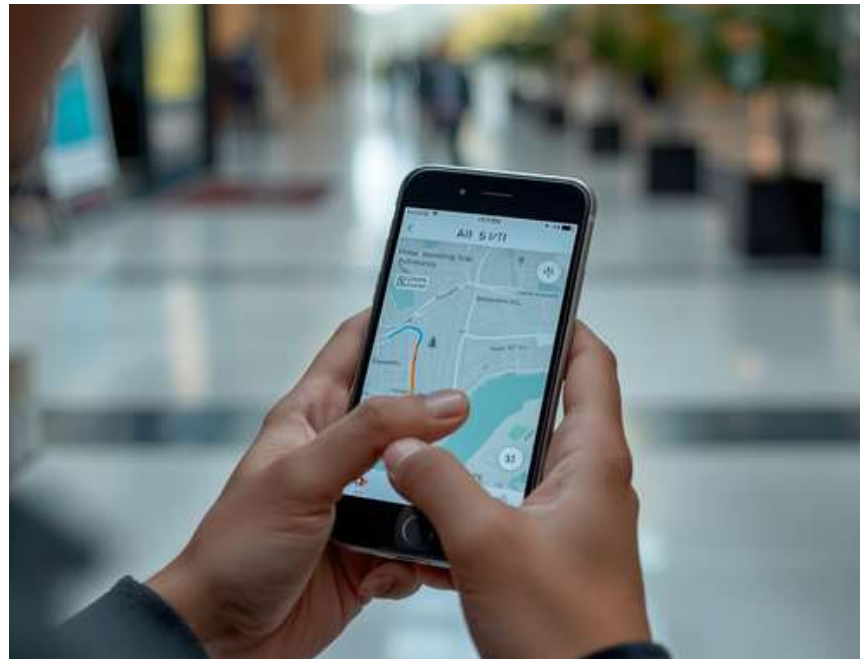
Existing System Architecture

Traditional navigation systems rely on static maps, physical signage, and manual directories. These systems lack real-time updates and cannot adapt to emergency situations. Key limitations include: no dynamic route recalculation, absence of emergency evacuation protocols, limited accessibility features, and dependency on pre-printed materials that quickly become outdated. The architecture typically consists of standalone kiosks or paper-based directories with no interconnected database or intelligent routing capabilities.

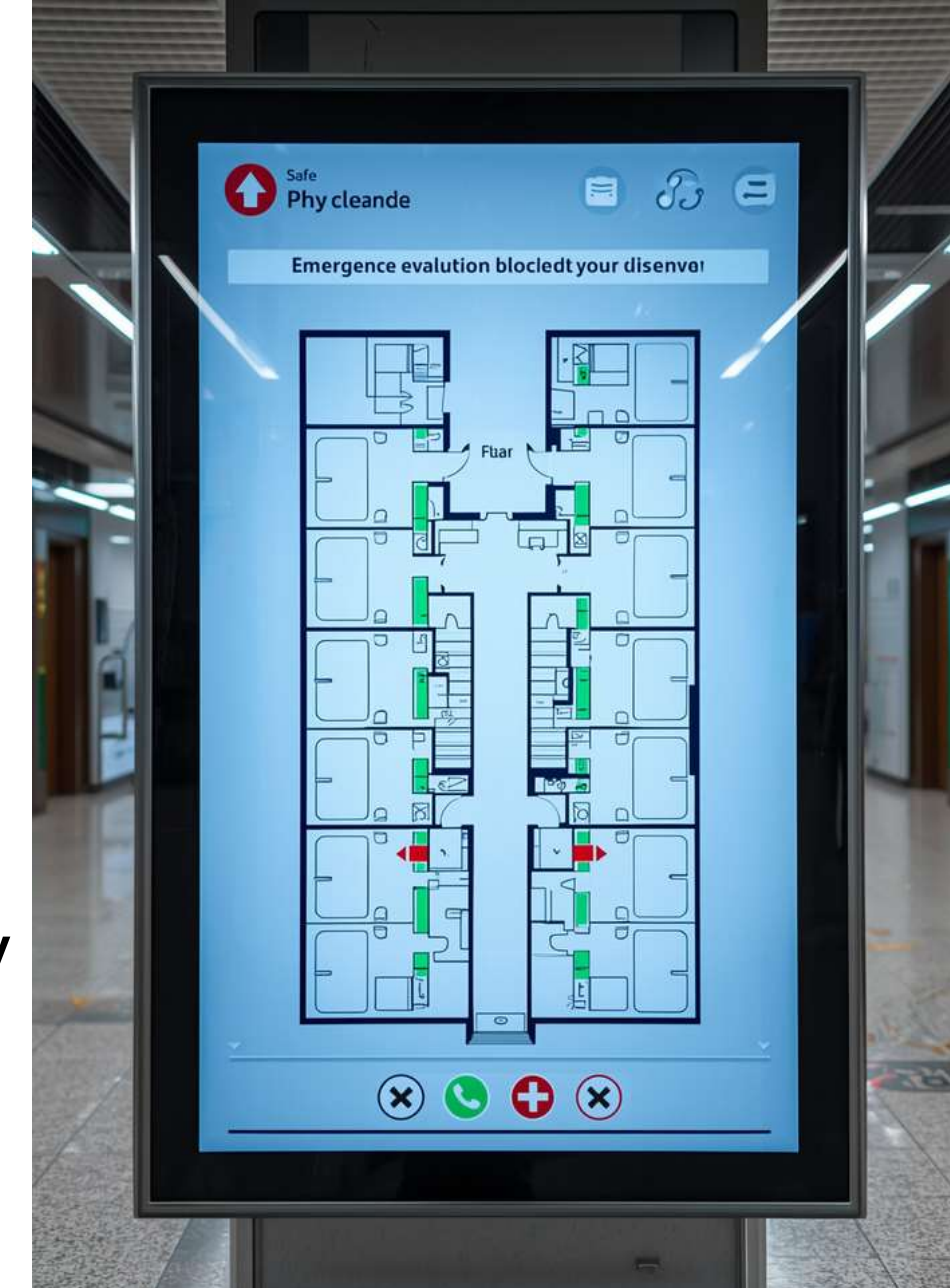


Proposed System

The Intelligent Navigation and Safety Platform introduces a revolutionary approach to indoor navigation and emergency response. Key features include real-time pathfinding algorithms that calculate optimal routes based on current conditions, dynamic emergency routing that automatically redirects users away from hazards, and location-based guidance using indoor positioning technology for accurate turn-by-turn directions.



Intelligent
Navigation Platform



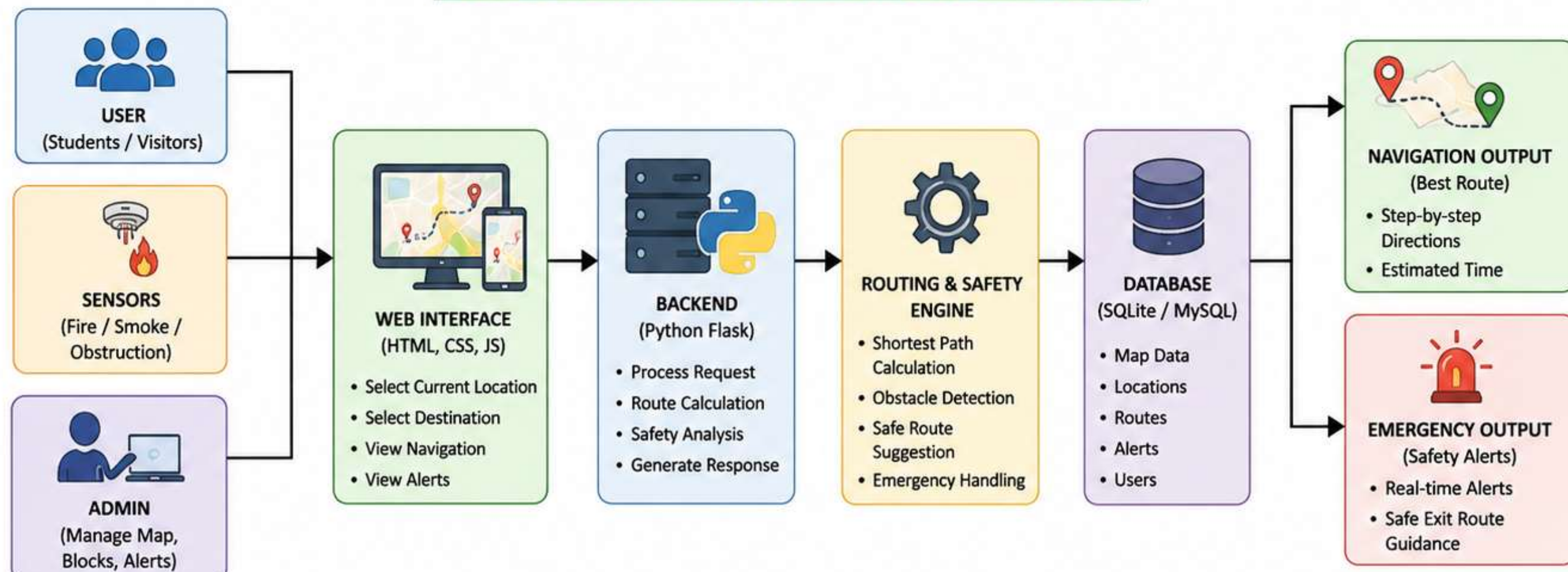
The system provides safe evacuation paths during emergencies by analyzing blocked routes and fire locations in real-time. Users receive instant notifications with alternative escape routes, while administrators can monitor evacuation progress through a centralized dashboard. Integration with building sensors enables automatic detection of obstacles, ensuring users always have access to the safest and fastest route to their destination or exit points.

EXISTING SYSTEM ARCHITECTURE



LIMITATIONS: No real-time navigation, No emergency alerts, No safe route guidance

PROPOSED SYSTEM ARCHITECTURE



The Challenge



Navigating Without Safety Intelligence

Current navigation systems prioritize speed over safety, leaving users unaware of real-time hazards, unsafe routes, and dangerous conditions that could put them at risk.





How It Works



Intelligent Route Analysis

SafePath's intelligent algorithms continuously analyze real-time traffic patterns, crime statistics, weather conditions, and community reports to calculate the safest route for your journey.



Thank You Navigate Safely with SafePath



+1-800-SAFEPATH



@safepath.platform



info@safepath.io | safepath.io



Key Benefits



Peace of Mind, 24/7

SafePath delivers intelligent protection for commuters, families, and travelers. With real-time safety updates and smart navigation, you're always connected to the safest routes and instant alerts.

